

TalentMatch

AI

Enterprise Workforce Intelligence

Problem & Solution

The Problem: Traditional recruiting relies on exact keyword matching, which is time-consuming and often misses the true semantic context of a candidate's skills.

The Solution: An intelligent matching engine powered by Vector Search and Machine Learning to understand context, identify gaps, and predict candidate success.

Talent Scout (Smart Matching)

- **Semantic Retrieval:** Utilizes ChromaDB vector embeddings for context-aware search.
- **Profile Alignment:** Evaluates candidates based on core technical competencies and experience.
- **Predictive Analytics:** Applies a Logistic Regression model to calculate a data-driven "Match Score".

TalentMatch.ai
ENTERPRISE HUB V2.0

- ☒ Talent Scout
- ☐ Team Builder
- ☐ Skill Gap Analysis
- ☐ AI Agent Chat

SYSTEM STATUS: ONLINE

Skill-Driven Talent Intelligence

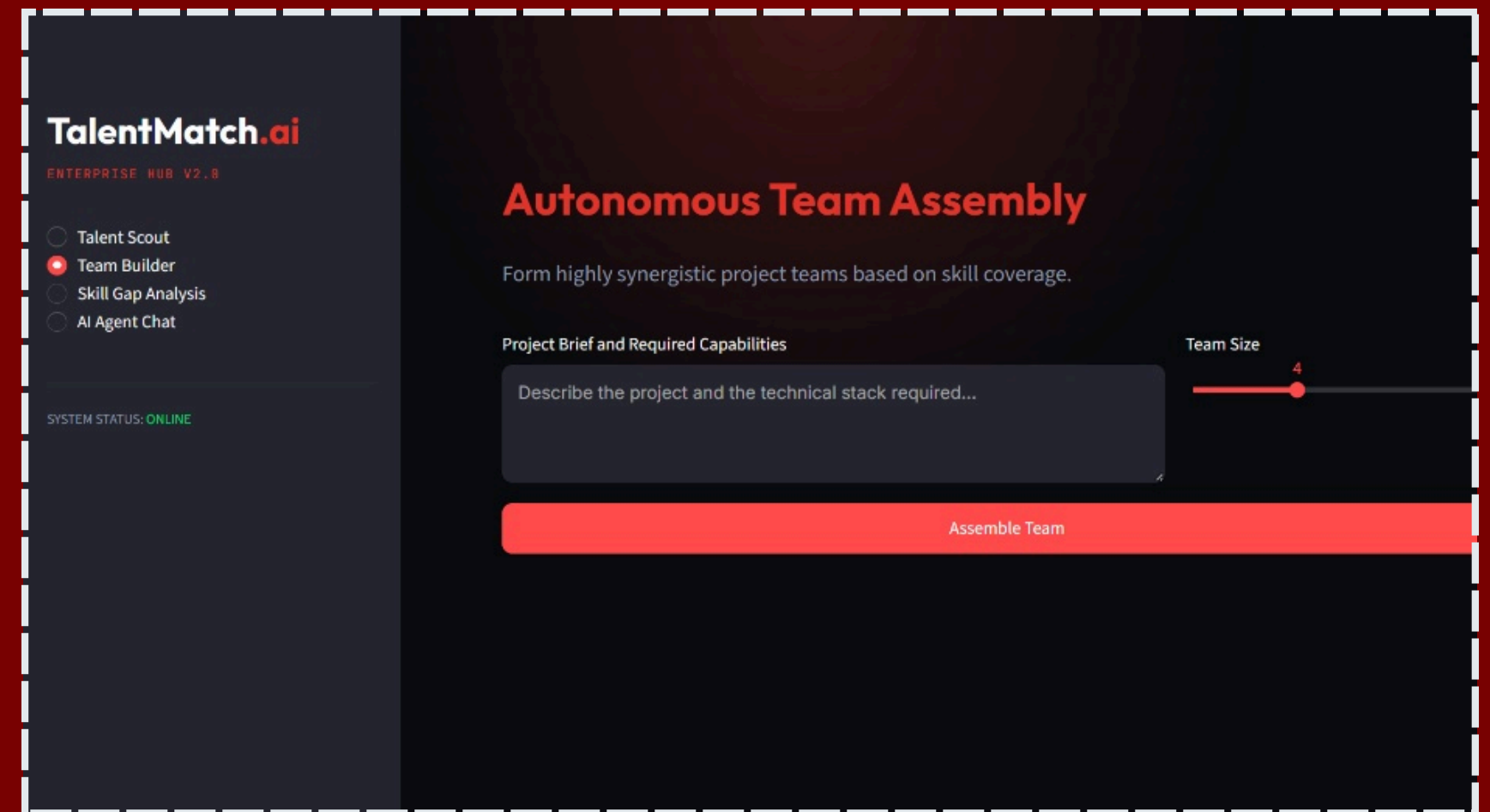
Search the vector index for optimal candidate alignment.

Required Technical Competencies

Find Matching Talent

Autonomous Team Assembly

- **Synergistic Assembly:** Automatically forms complementary project teams.
- **Maximum Coverage:** Optimizes team selection to ensure full required skill coverage.
- **Actionable Insights:** Provides AI-generated recommendations to address any remaining team weaknesses.



Skill Gap Analysis

- **Role Profiling:** Compares current employee proficiencies against target role requirements.
- **Deficit Identification:** Accurately isolates critical capability gaps.
- **Targeted Upskilling:** Auto-recommends specific training modules to bridge the identified gaps immediately.



The screenshot displays the TalentMatch.ai interface for Skill Gap Analysis. The left sidebar contains the company logo, version information, a navigation menu with 'Skill Gap Analysis' selected, and a system status indicator. The main panel features a title, a description, input fields for Employee ID and Target Role, and an 'Analyze Profile' button.

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Skill Gap Analysis

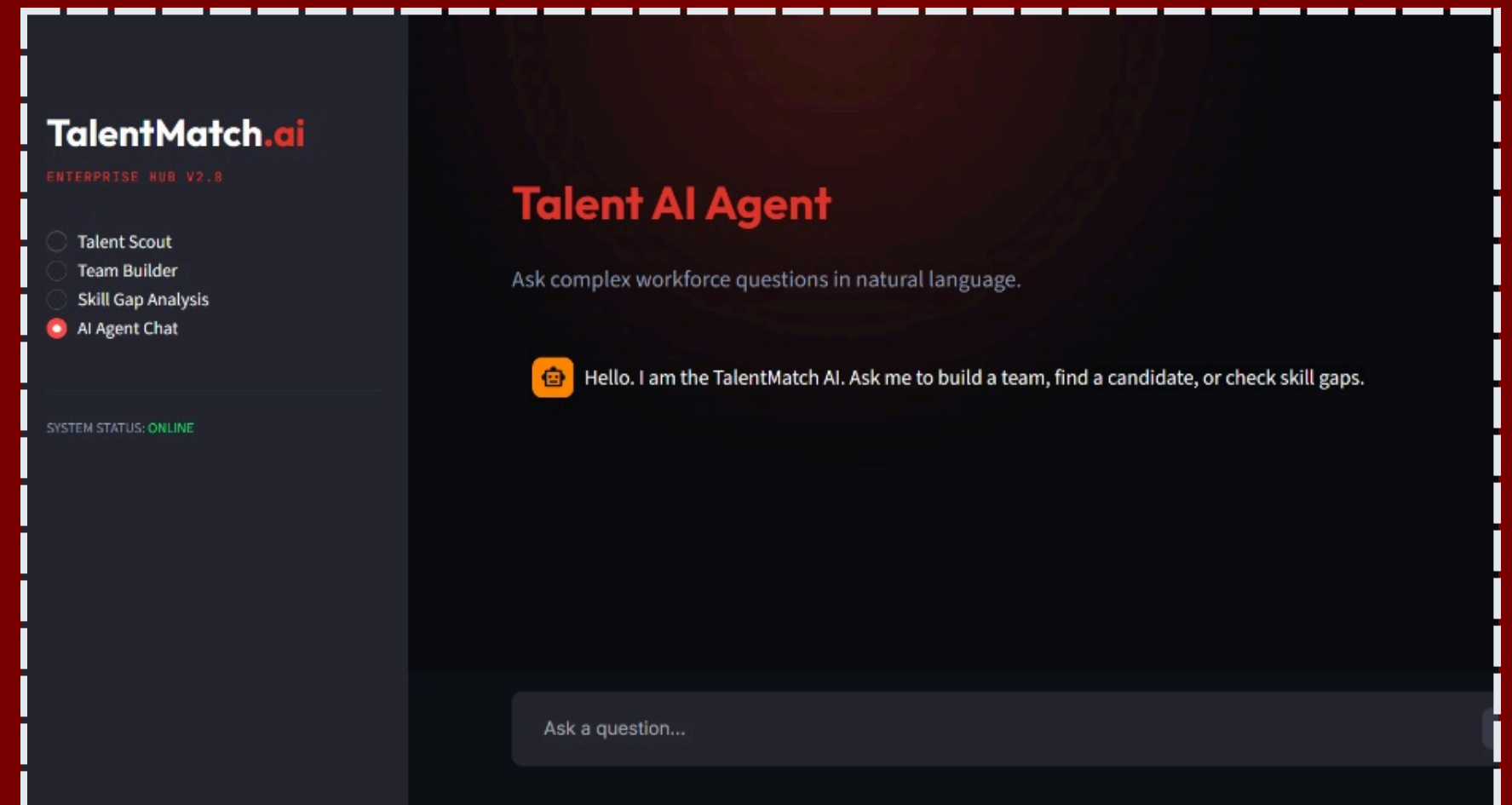
Identify capability deficits and auto-recommend training.

Employee ID:

Target Role:

Talent AI Agent

- **Conversational UI:** A Natural Language Interface for complex workforce inquiries.
- **Smart Routing:** Intelligently analyzes prompts and invokes appropriate system tools automatically.
- **Explainable AI:** Generates responses backed by traceable data records and execution logs.



System Architecture & Tech Stack



Frontend

Streamlit



Backend API

FastAPI



Machine Learning

Scikit-learn



Vector Database

ChromaDB



Relational Database

PostgreSQL